



PHYSICS HSSC-II

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

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Deleting/overwriting is not allowed. Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. If the distance between the centers of two identical bodies is doubled, the gravitational force 'F' between them will become:	F	2F	$\frac{F}{2}$	$\frac{F}{4}$	☐	☐	☐	☐
2. If the gravitational field strength at the earth's surface is 'g'. What is its value at a height, $h=R_e$ from surface of earth?	$\frac{g}{2}$	$\frac{g}{4}$	2g	$\frac{g}{9}$	☐	☐	☐	☐
3. Two stars have the same luminosity. If one appears dimmer, it is because:	It is farther from the observer	It has lower temperature	It has smaller size	It emits less energy	☐	☐	☐	☐
4. In SHM, at which value of θ , the instantaneous velocity(v) is maximum (v_0)?	$\theta = \frac{3\pi}{4}$	$\theta = \frac{\pi}{2}$	$\theta = \frac{\pi}{4}$	$\theta = 0$	☐	☐	☐	☐
5. Total energy of the mass spring system in SHM, is proportional to:	$\frac{m\omega}{x_0^2}$	$\frac{mx_0}{\omega^2}$	$m\omega^2 x_0^2$	$m\omega x_0$	☐	☐	☐	☐
6. If the number of lines per unit length in a grating increase, the diffraction pattern (maxima) becomes:	Less sharp and closely spaced	Sharper and widely spaced	More widely spaced and less sharp	More closely spaced and sharper	☐	☐	☐	☐
7. A.C voltage across a $0.5\mu F$ capacitor is $16\sin(2\times 10^3 t)V$, peak value of voltage through the capacitor is:	2×10^3	10^3	16	2	☐	☐	☐	☐
8. A simple pendulum has a period of 2.5s. What will be its period if its length becomes four times large?	2.5	1.25s	0.625s	5s	☐	☐	☐	☐
9. A bridge rectifier is used to:	Increase voltage	Convert AC to DC	Store charge	Convert DC to AC	☐	☐	☐	☐
10. In a purely resistive AC circuit, the phase difference between voltage and current is:	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	☐	☐	☐	☐
11. In photoelectric effect, which factor increases by increasing the intensity of incident light?	Kinetic energy of electron	Stopping potential	Number of emitted electrons	Work function	☐	☐	☐	☐
12. If a particle of mass m is moving with speed v then De-Broglie wavelength λ is given by:	$\lambda = \frac{3h}{mv}$	$\lambda = \frac{h}{mv}$	$\lambda = \frac{2h}{mv}$	$\lambda = \frac{h}{2mv}$	☐	☐	☐	☐

Question	A	B	C	D	A	B	C	D
13. Which of the following is an example of nuclear fusion?	Uranium decay	Atomic bomb	Energy production in the Sun	Radioactive decay	☐	☐	☐	☐
14. If the distance from a star (luminosity L) is doubled, the radiant flux intensity becomes:	One-fourth	Half	Double	Four times	☐	☐	☐	☐
15. The magnitude of Coriolis Force (F_c) that bends the air packet (mass= $2 \times 10^5 \text{ kg}$) (velocity 22.2 ms^{-1}) at 25° is:	2.74 N	274 N	$2.74 \times 10^4 \text{ N}$	$2.74 \times 10^6 \text{ N}$	☐	☐	☐	☐
16. If a star is receding from the earth it exhibits:	Red shift	Blue shift	Green shift	Yellow light	☐	☐	☐	☐
17. Which color of visible light has the lowest frequency?	Red	Green	Blue	Violet	☐	☐	☐	☐

—2HA-I 26004- 40041 (B) —

$$F = \frac{Gm_1m_2}{r^2}$$

$$T = \frac{1}{f}$$

$$F = \frac{L}{A}$$

$$c = \frac{\epsilon_0 A}{d}$$

$$\lambda = \frac{h}{p}$$

$$T = 2\pi \sqrt{\frac{\ell}{g}}$$

$$a = -\omega^2 x$$

$$p = mv$$

$$h = 6.63 \times 10^{-34} \text{ Js}^{-1}$$

$$R = R_e + h$$

$$g = 9.8 \text{ ms}^{-2}$$

$$rV = kq$$

$$v = v_o \sin(\theta)$$

$$\theta = \omega t$$

$$\omega_{earth} = 7.3 \times 10^{-5} \text{ rads}^{-1}$$

$$F_c = 2mv\omega_{earth} \sin \theta$$

$$E = hf$$

$$P = \frac{E}{C}$$

$$g = \frac{GM_e}{R^2}$$

$$V = \frac{W}{Q}$$

ROLL NUMBER					



PHYSICS HSSC-II

26

Time allowed: 2:35 Hours

Total Marks Sections B and C: 68

SECTION – B (Marks 42)

Q. 2 Answer the following parts briefly.

(14x3=42)

(i)	When a satellite is put into a higher orbit what will be the effect on its orbital velocity? Explain mathematically.	1+2	OR	Time period of a mass attached at the end of a spring is 0.4s. Find out the magnitude of acceleration when the displacement is 4cm.	03
(ii)	What happens to pressure when temperature and molecular speed of gas molecules increase?	03	OR	Briefly explain constructive and destructive interference of light. Give mathematical expression for both in terms of path difference.	1.5 + 1.5
(iii)	Justify qualitatively the factors which determine the frequency response and sharpness of resonance.	03	OR	Explain the function of capacitor in bridge rectifier circuit.	03
(iv)	What is self-inductance? Also write its SI unit and the factors upon which it depends.	1+2	OR	Find field strength 'g' at a height of 1500 km above surface of Earth.	03
(v)	Derive the relation between absolute temperature and average kinetic energy of gas molecules.	03	OR	How does doubling the frequency affect the capacitive reactance? Show mathematically.	03
(vi)	Briefly explain the significance of wind pattern on climate system.	1x3	OR	How are X-rays used in imaging internal body structure? Explain briefly.	03
(vii)	What is meant by pair production? Also calculate the minimum energy required in this process.	1+2	OR	Briefly explain the two types of climate inertia.	1.5 + 1.5
(viii)	What is meant by Piezoelectric effect? Enlist its two applications.	1+2	OR	What is the role of electron wavelength in achieving high resolution of electron microscope?	03
(ix)	How do control rods and moderator play an important role in the action of nuclear reactor?	1.5 + 1.5	OR	Draw labelled wave form and phasor diagram between applied voltage and current for pure inductor A.C.	2+1
(x)	Why does the equatorial region receive more energy than polar regions?	03	OR	To which idea does a red-shift exhibited by a star leads and why?	1+2
(xi)	Calculate root mean square speed of oxygen molecule at standard temperature and pressure.	03	OR	Briefly explain the terms: a. Mass Defect b. Binding Energy	1.5 + 1.5
(xii)	What factors determine the strength of gravitational field 'g' around a planet?	1.5 + 1.5	OR	What is meant by light damping and heavy damping? Show these graphically. (Displacement-time graph)	1x3
(xiii)	What is meant by grating element of a diffraction grating? Also write its formula.	2+1	OR	Distinguish graphically between half-wave and full-wave rectification.	1.5+ 1.5
(xiv)	How does increasing slit separation affect fringe spacing? Discuss with mathematical relation.	2+1	OR	Four capacitors of capacitance 'C' are connected as shown. Calculate their equivalent capacitance.	03

SECTION – C (Marks 26)

Note: Attempt the following questions.

Q.3	Briefly describe gravitational potential and derive the expression $\phi = -\frac{GM}{r}$ at a distance r from the centre of Earth. Make a sketch.	2+4 +1	OR	What is meant by simple harmonic motion (SHM)? Show that mass attached to a spring executes SHM. Derive the expressions for frequency 'f' and time period 'T'. Draw diagram as well.	1+3 +2+1
Q.4	Explain the working of Young's double slit experiment. Make diagram. Also derive the expression for fringe spacing.	4+1 +2	OR	What is meant by electric potential? Derive an expression for electric potential in the field due to a point charge. Make sketch.	1+5 +1
Q.5	In an RL series circuit, will the current lag or lead the applied alternating voltage? Explain the answer with phasor diagram. Also find an expression for the impedance of RL series circuit.	2+2 +2	OR	What is meant by photo electric effect? Why no emission of electrons occurs below threshold frequency? Discuss the effects of intensity of light on photoelectric current with graph.	1+2 +3
Q.6	a. What is meant by Radiant Flux Intensity? How it follows inverse square law? b. A star has a known luminosity of $9.7 \times 10^{27} W$. Observations of the star show that the radiant flux intensity of light received on Earth from the star is $114 nW m^{-2}$. Determine the distance of the star from Earth.	3+3	OR	a. What is meant by enrichment? Why Uranium fuel needs to enriched before use? b. Find decay constant and activity of an element if its 100 out of 700 nuclei decay in 5 min.	(2+1) +3

— 2HA-I 26004 (B) —

$$\begin{array}{llllllll}
 v_o = \sqrt{\frac{GM_e}{R}} & \omega = \sqrt{\frac{k}{m}} & X_c = \frac{1}{\omega C} & V_{rms} = \frac{V_o}{\sqrt{2}} & g = G \frac{M_e}{R^2} & I_{rms} = \frac{I_o}{\sqrt{2}} & T = \frac{2}{3k} \left(\frac{1}{2} m v^2 \right) & m = \frac{M}{N_A} \\
 F = kx & \omega = 2\pi f & M_e = 6 \times 10^{24} kg & a = -\omega^2 x & R_e = 6.4 \times 10^6 m & N_A = 6.022 \times 10^{23} & A = \lambda N & M_{oxygen} = 32g \\
 F = \frac{L}{A} & T_{\frac{1}{2}} = \frac{0.693}{\lambda} & \lambda = \frac{1}{N} \times \frac{\Delta N}{\Delta t} & v_{rms} = \sqrt{\frac{3P}{\rho}} & v_{rms} = \sqrt{\frac{3kT}{m}} & v_{rms} = \sqrt{\frac{3RT}{M}} & f = \frac{1}{T} & F = \frac{L}{4\pi d^2} \\
 k = 1.38 \times 10^{-23} JK^{-1} & R = 8.314 Jmol^{-1} K^{-1} & & R = R_e + h & \sigma = 5.67 \times 10^{-8} Wm^{-2} K^{-4} & & &
 \end{array}$$



PHYSICS HSSC-II

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Signature should not cross this line

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Deleting/overwriting is not allowed. Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Velocity of a wind packet (mass = $2 \times 10^5 \text{ kg}$), if a Coriolis Force (F_c) of 274 N bends it at an angle of 25° would be:	32.2 km hr^{-1}	42.2 km hr^{-1}	12.2 ms^{-1}	22.2 ms^{-1}	☐	☐	☐	☐
2. Which of the following electro-magnetic waves has the highest frequency?	Gamma rays	Ultraviolet	Microwaves	Infrared	☐	☐	☐	☐
3. Two objects attract each other with a gravitational force 'F'. If mass of one object is doubled and distance between their centers remains same, then gravitational force is:	2F	4F	F/2	F	☐	☐	☐	☐
4. The gravitational field strength at a point depends on:	Shape of the test body	Velocity of the body	Mass of the test body	Mass causing the field	☐	☐	☐	☐
5. The SI unit of luminosity 'L' is:	Watt m^{-2}	Kelvin	Watt	Joule	☐	☐	☐	☐
6. In SHM, the relationship between acceleration and displacement of a particle when $\omega = 10$ is:	$a = -100x$	$a = 100x^2$	$a = 10x$	$a = -10x^2$	☐	☐	☐	☐
7. What is the frequency of the body when its time period is 2s?	0.5 Hz	0.2 Hz	1 Hz	2 Hz	☐	☐	☐	☐
8. The wavelength of light can be accurately determined by using a:	Interferometer	Capacitor	Melde's apparatus	Diffraction grating	☐	☐	☐	☐
9. Electric potential due to a point charge varies with distance as:	$1/r^2$	r^2	$1/r$	r	☐	☐	☐	☐
10. The capacitance of a parallel plate capacitor increases when:	Area of plates increases	Using battery of low potential	Distance between plates increases	Area of plates decreases	☐	☐	☐	☐
11. In a bridge rectifier, current flows through the load:	During both half cycles in the same direction	During both half cycles in alternating directions	Only during positive half cycle	Only during negative half cycle	☐	☐	☐	☐
12. In a pure inductive AC circuit, the applied voltage leads the current by:	0	$\frac{\pi}{2}$	$\frac{3\pi}{2}$	π	☐	☐	☐	☐

Question	A	B	C	D	A	B	C	D
13. De-Broglie wavelength of a body having mass 0.5 kg moving with a speed of 36 km/hr is:	$1.3 \times 10^{-34} \text{ mm}$	$1.3 \times 10^{-24} \text{ mm}$	$1.3 \times 10^{-24} \text{ m}$	$1.3 \times 10^{-34} \text{ m}$	☐	☐	☐	☐
14. Number of electrons emitted in photoelectric effect depends upon:	Energy of incident light	Wavelength of incident light	Intensity of Incident light	Frequency of incident light	☐	☐	☐	☐
15. In nuclear fission, a chain reaction occurs due to the release of:	Photons	Neutrons	Protons	Electrons	☐	☐	☐	☐
16. According to the inverse square law, radiant flux intensity is inversely proportional to:	Temperature	Luminosity	Distance	Square of distance	☐	☐	☐	☐
17. Which layer of the atmosphere contains the ozone layer?	Mesosphere	Stratosphere	Troposphere	Thermosphere	☐	☐	☐	☐

—2HA-I 26004-42043 (D) —

$$F = \frac{Gm_1m_2}{r^2}$$

$$T = \frac{1}{f}$$

$$F = \frac{L}{A}$$

$$c = \frac{\epsilon_0 A}{d}$$

$$\lambda = \frac{h}{p}$$

$$T = 2\pi \sqrt{\frac{\ell}{g}}$$

$$a = -\omega^2 x$$

$$p = mv$$

$$h = 6.63 \times 10^{-34} \text{ Js}^{-1}$$

$$R = R_e + h$$

$$g = 9.8 \text{ ms}^{-2}$$

$$rV = kq$$

$$v = v_o \sin(\theta)$$

$$\theta = \omega t$$

$$\omega_{\text{earth}} = 7.3 \times 10^{-5} \text{ rads}^{-1}$$

$$F_c = 2mv\omega_{\text{earth}} \sin \theta$$

$$E = hf$$

$$P = \frac{E}{C}$$

$$g = \frac{GM_e}{R^2}$$

$$V = \frac{W}{Q}$$

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PHYSICS HSSC-II

28

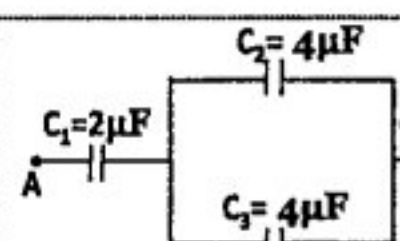
Time allowed: 2:35 Hours

Total Marks Sections B and C: 68

SECTION – B (Marks 42)

Q. 2 Answer the following parts briefly.

(14x3=42)

(i)	How will the orbital velocity of a satellite be affected when it is put into a lower orbit (closer to the Earth)? Explain mathematically.	1+2	OR	A body with mass 800g attached to a spring executes SHM. The restoring force is 60N when the displacement is 0.3m. a. Find out its angular frequency b. Also find the magnitude of acceleration when the displacement is 12cm.	1+1+1
(ii)	How is the pressure related to the density of the gas? Also give its mathematical expression.	2+1	OR	State the conditions necessary for observing interference fringes.	1x3
(iii)	How does resonance play an important role in tuning of radio station?	03	OR	Briefly explain the use of capacitors in any three household appliances.	1x3
(iv)	What is mutual inductance? Also write its SI unit and the factors upon which it depends.	1+2	OR	What will be the value of gravitational field strength 'g' at 35,700 km, where a satellite is orbiting the earth?	03
(v)	What is meant by breaking radiation? Also write the relation for minimum wavelength of x-rays.	2+1	OR	An AC voltage across a $0.5\mu\text{F}$ capacitor is $16\sin(2 \times 10^3 t)\text{V}$. Find its capacitive reactance.	03
(vi)	Explain the role of atmosphere in maintaining climate balance.	03	OR	What is meant by contrast in X-ray imaging?	03
(vii)	Briefly describe pair annihilation. Also justify the conservation of energy and momentum in this process.	1+2	OR	How does ocean play a main role in climate inertia?	03
(viii)	Briefly discuss two applications of ultrasound in medical technology.	03	OR	Why does electron microscope have higher resolution than optical microscope? (Compare both.)	1.5+1.5
(ix)	How is the variation of binding energy per nucleon an important factor in nuclear physics? (Discuss both high binding energy and low binding energy)	1.5+1.5	OR	An AC circuit consists of pure resistance of 20Ω and is connected across AC supply of 220V, 50Hz. Calculate: a. Peak value of voltage. b. Peak value of current.	03
(x)	How does energy imbalance between equator and poles affect earth climate?	03	OR	How can Wien's displacement law be used to determine the temperature of a star? Also write its formula.	2+1
(xi)	Why the value of 'g' does not change significantly near the Earth's surface? (Show mathematically)	03	OR	Briefly explain the function of control rods and moderators in nuclear reactor.	1.5+1.5
(xii)	Calculate the average translational kinetic energy of a gas molecule at a temperature 47°C .	03	OR	What is meant by free oscillations? Also give its two examples.	1+2
(xiii)	How does increasing the number of slits per unit length affect the diffraction pattern? Make a sketch.	2+1	OR	Distinguish between peak value and root mean square (rms) values of alternating voltage.	1.5+1.5
(xiv)	How can one increase the fringe spacing in Young's double slit experiment? Also write its mathematical calculation.	2+1	OR	Deduce the equivalent capacitance between A and B for the given circuit. 	03

SECTION – C (Marks 26)

Note: Attempt the following questions.

Q.3	Briefly describe gravitational potential and derive the expression $\phi = -\frac{GM}{r}$ at a distance r from the center of Earth. Make a sketch.	2+4+1	OR	Derive expressions for Kinetic and Potential energies of a mass attached to a spring executing SHM. Show that total energy of the spring system remains constant. Also make neat sketch.	2+2+2+1
Q.4	Explain the working of Young's double slit experiment. Make diagram. Also derive the expression for fringe spacing.	4+1+2	OR	a. Determine the electrical potential energy stored in a capacitor by using graph. b. Calculate the energy stored, in a capacitor having 5C of charge and potential different of 15V.	(3+1)+3
Q.5	Explain the working of a bridge rectifier using four diodes. Also draw the circuit diagram.	4+2	OR	What is photoelectric effect? Explain work function of metal and Einstein photo electric equation. What is the effect of frequency of radiation on K.E of photoelectrons?	1+3+2
Q.6	a. What is meant by luminosity of a star? Write the relation between luminosity, radius and temperature. b. What are the relative luminosities of the objects A and B if $T_A=100\text{K}$, $T_B=200\text{K}$, $R_A=10\text{m}$ and $R_B=5\text{m}$	3+3	OR	a. Derive and show that the decay of nucleus is exponential with time. b. If half-life of strontium (Sr. 91) is 9.70 hours. Find its decay constant.	3+3

— 2HA-I 26004 (D) —

$$\begin{array}{llllllll}
 v_o = \sqrt{\frac{GM_e}{R}} & \omega = \sqrt{\frac{k}{m}} & X_c = \frac{1}{\omega C} & V_{rms} = \frac{V_o}{\sqrt{2}} & g = G \frac{M_e}{R^2} & I_{rms} = \frac{I_o}{\sqrt{2}} & T = \frac{2}{3k} \left(\frac{1}{2} m v^2 \right) & m = \frac{M}{N_A} \\
 F = kx & \omega = 2\pi f & M_e = 6 \times 10^{24} \text{ kg} & a = -\omega^2 x & R_e = 6.4 \times 10^6 \text{ m} & N_A = 6.022 \times 10^{23} & A = \lambda N & M_{oxygen} = 32 \text{ g} \\
 F = \frac{L}{A} & T_{1/2} = \frac{0.693}{\lambda} & \lambda = \frac{1}{N} \times \frac{\Delta N}{\Delta t} & v_{rms} = \sqrt{\frac{3P}{\rho}} & v_{rms} = \sqrt{\frac{3kT}{m}} & v_{rms} = \sqrt{\frac{3RT}{M}} & f = \frac{1}{T} & F = \frac{L}{4\pi d^2} \\
 k = 1.38 \times 10^{-23} \text{ JK}^{-1} & & R = 8.314 \text{ Jmol}^{-1} \text{ K}^{-1} & & R = R_e + h & \sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2} \text{ K}^{-4} & &
 \end{array}$$